

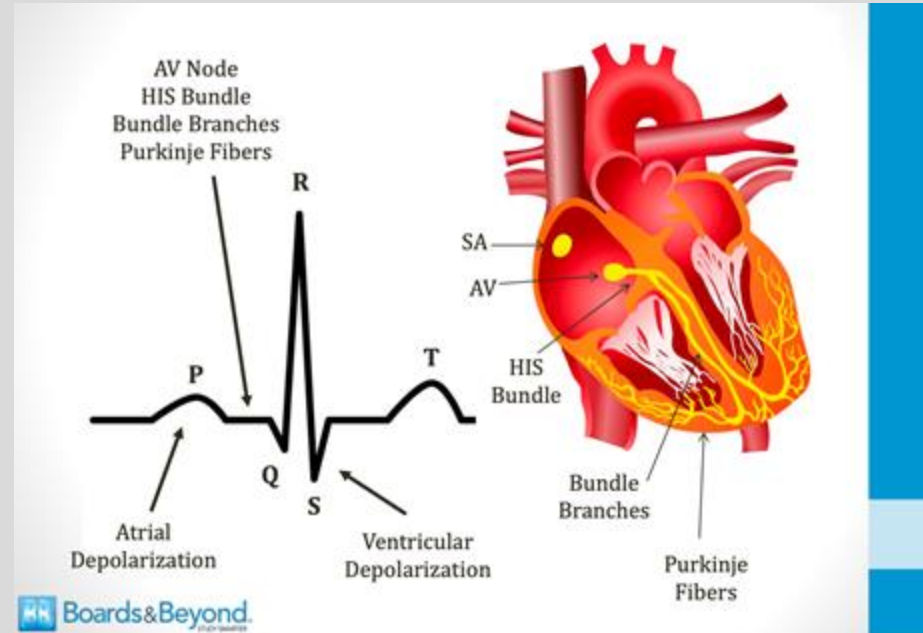
Session Objectives

1. Differentiate and manage the different types of AV blocks.
2. Understand bundle anatomy and identify between left vs right bundle branch block.
3. Recognize ECG changes due to electrolyte imbalances.

Heart or AV Conduction Blocks (same thing)

- Delay or interruption in the transmission of the impulse from the atria to the ventricles
- Two locations: AV node and His-Purkinje system

Disorder	Common Site of Block
1 st Degree	AV node
Mobitz I	AV Node
Mobitz II	His-Purkinje System
3 rd degree (Complete)	His-Purkinje System



First Degree AV Heart Block

- Delayed conduction from atrium to ventricle (but no interruption)
- **Prolonged PR interval** (> 200 msec)
- All PR intervals are equal and all P waves Conducted (QRS follows)
- Usually **asymptomatic**
- Can be caused by medications
- **No treatment required**



Second Degree AV Block: Mobitz Type I (Wenckebach)

- The **PR interval progressively increases** until one p-wave is blocked and isn't transmitted to the ventricles
- Notable **PR interval lengthening with a P-wave not followed by a QRS complex**
- The cycle starts all over again after that drop (regularly irregular)
- **Treatment is usually not indicated** as this rhythm usually is asymptomatic



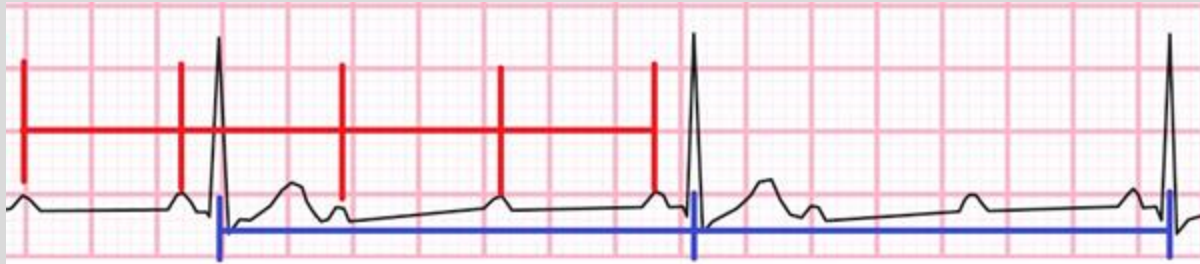
Second Degree AV Block: Mobitz Type II

- Dropped QRS complex that is **NOT** preceded by a lengthening of the PR interval
- The **PR interval is consistent** (either normal or prolonged), until one p-wave is blocked and isn't transmitted to the ventricles
- Symptoms may be present and are more common than Mobitz Type I and include presyncope, syncope, fatigue, or SOB
- More serious than type I second-degree AV block
- May progress to third degree (complete) AV block
- Often **treated with a pacemaker**

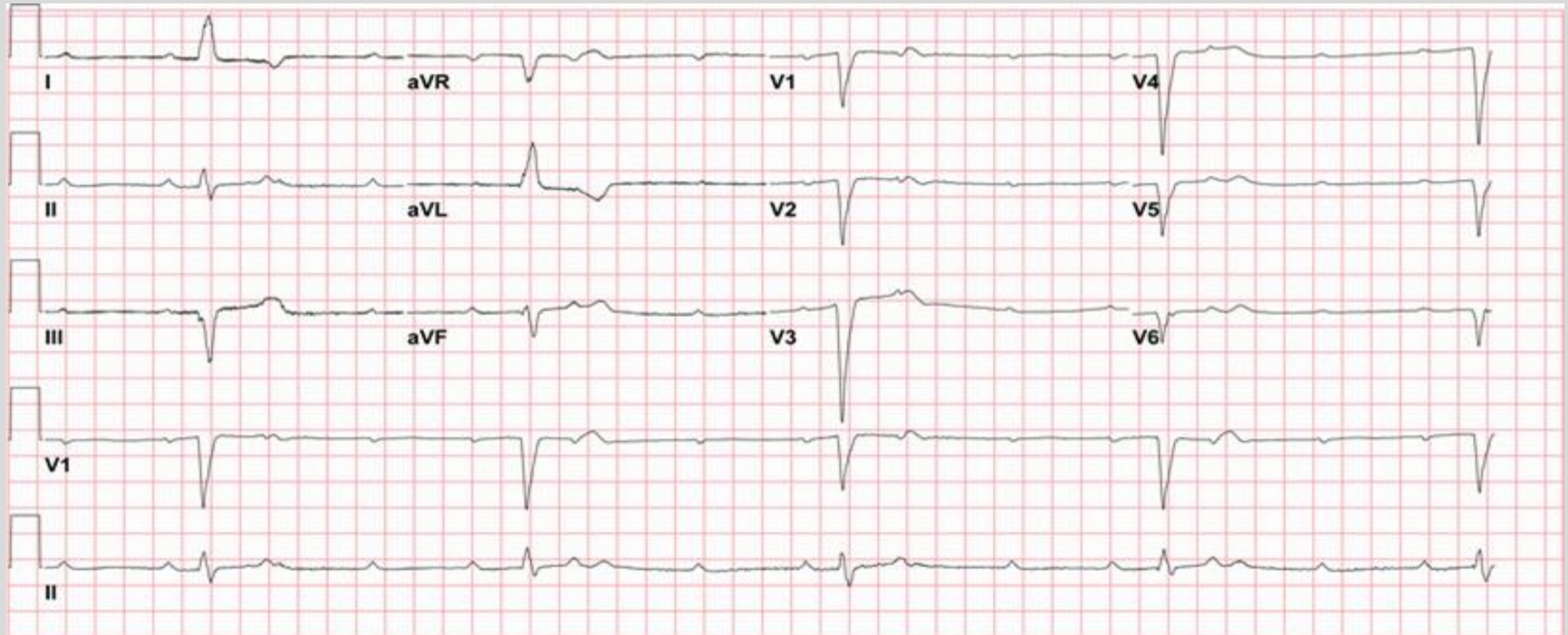


Third Degree (Complete) Heart Block

- The **atria and ventricles beat independently** of each other
- P-waves are **not associated** with QRS complexes
- **Atrial pacing is greater than ventricular pacing**
- ECG will show **equal RR intervals** and **equal PP intervals** but no association
- May be caused by MI, degeneration of the conductive tissue, and **Lyme Disease**
- Treated with a **Pacemaker**



Third Degree (Complete) Heart Block



Summary: AV Blocks

Site of Block:*

AV node

AV node

His-Purkinje

His-Purkinje

First degree AV block



Second degree AV block (Mobitz I or Wenckebach)



Second degree AV block (Mobitz II)



Third degree AV block



The Heart Block Poem

- If the R is far from the P, then you have a 1st Degree



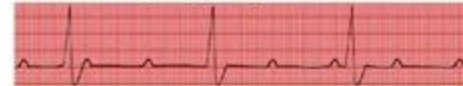
- PR gets longer, longer, longer, drop, it's a case of Wenckebach!



- If some R's don't get through, prepare to pace that Mobitz III!



- If the R's & P's don't agree, prepare to PACE that 3rd degree!

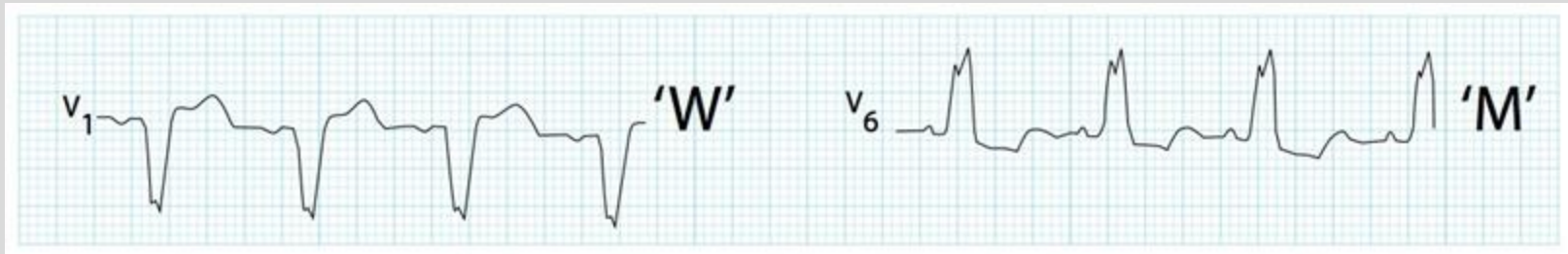


Heart Block Poem

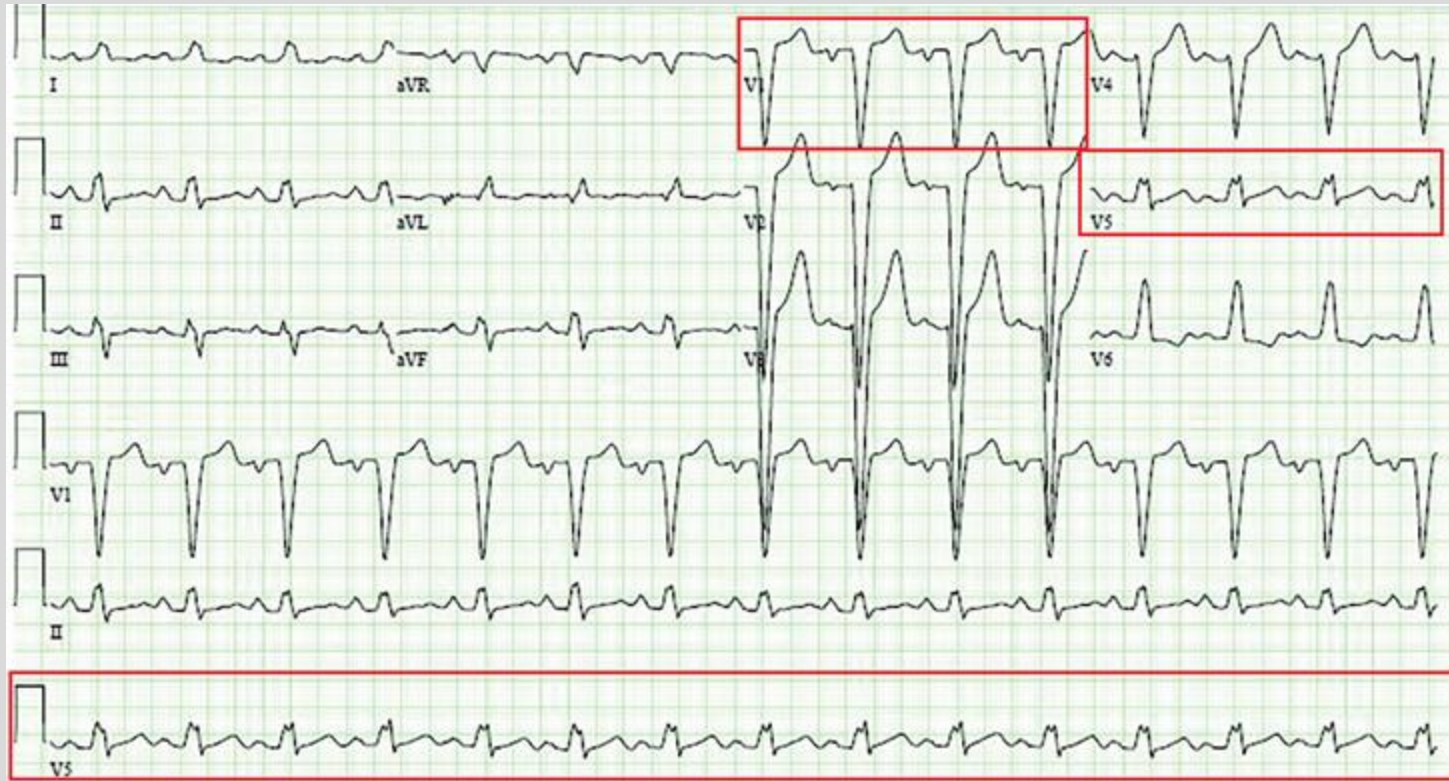
*Least to most severe

Left Bundle Branch Block (LBBB)

- Normally, the septum is activated from left to right
- In LBBB, septal depolarization is **reversed**
- Leads to **QRS duration > 120ms**
 - Produces tall R waves in the lateral leads (I, V5-6) and deep S waves in the right precordial leads (V1-3)
- Produces a broad or notched ('M'-shaped) R wave in the lateral leads (V5, V6) and ('W'-shaped) dominant deep S waves in V1

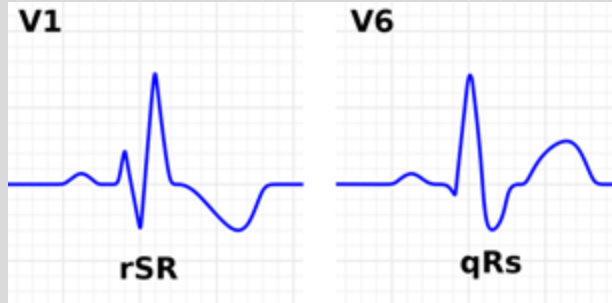


Left Bundle Branch Block (LBBB)



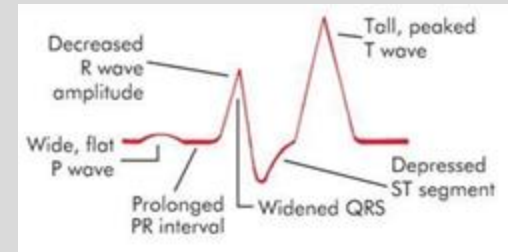
Right Bundle Branch Block (RBBB)

- QRS duration is greater than 120 msec
- QRS in **V1-V2 displays rSR' pattern** “bunny ears” or the letter “M”
- Lateral leads (I, V6) broad deep S-wave that is greater than R-wave duration.



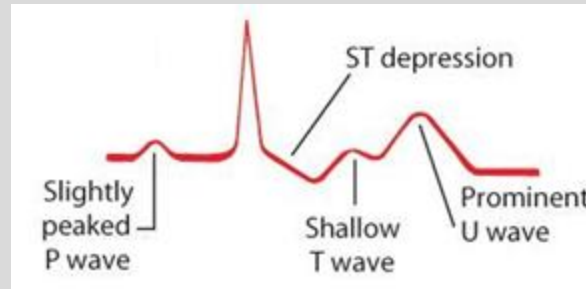
Hyperkalemia

- Potassium (K⁺): 3.5-5.0 mEq/L
- ECG changes depend on serum level and how rapid of a change
- K⁺ = 5.5 - 6.5 mEq/L
 - ****Tall, peaked, narrow based T waves****
 - QT interval shortening
- K⁺ = 6.5 - 7.5 mEq/L
 - First-degree AV block
 - **QRS widening**
- K⁺ > 7.5 mEq/L
 - Disappearance of P waves



Hypokalemia (HypUkalemia)

- Severe or symptomatic hypokalemia
 - Depression of the ST segment, decrease in the amplitude of the T-wave, and an increase in the amplitude of U waves
 - U-waves are often seen in the lateral precordial leads V4 to V6
 - Prolongs the QT interval
 - Increased amplitude and width of the P-wave



Summary

- Hyperkalemia: tall peaked T-waves, QT interval shortening
- Hypokalemia: ST segment depression, shallow T-waves, U-waves
- Hypercalcemia: QT interval shortening
- Hypocalcemia: Prolonged QT interval
- Hypothermia: Osborne “J” waves

Sick Sinus Syndrome

- When the SA node is damaged and no longer generates normal heartbeats at a normal rate
 - Another cause of Pathologic Bradycardia
- Most common caused is gradual loss of SA node function that comes with age
- Sx: Dizziness, Pre-syncope, Syncope, SOB, Fatigue
- Tx: Pacemaker